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EFFECTS OF INFORMATION TECHNOLOGY FAILURES ON THE MARKET VALUE OF FIRMS

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THE EFFECTS OF INFORMATION TECHNOLOGY FAILURES ON THE MARKET VALUE OF FIRMS

ABSTRACT

IT failures abound but little is known about the financial impact that these failures have on a firm's market value. Using the resource-based view of the firm and event study methodology, this study analyzes how firms are penalized by the market when they experience unforeseen operating or implementation-related IT failures. Our sample consists of 213 newspaper reports of IT failures by publicly traded firms, which occurred during a ten-year period. The findings show that IT failures result in a 2% average cumulative abnormal drop in stock prices over a two-day event window. The results also reveal that the market responds more negatively to implementation failures affecting new systems than to operating failures involving current systems. Further, the study demonstrates that more severe IT failures result in a greater decline in firm value and that firms with a history of IT failures suffer a greater negative impact. The implications of these findings for research and practice are discussed.

INTRODUCTION

In the summer of 1993, Greyhound Lines rolled out a new reservation system, Trips. The system turned out to be slow and crash-prone, impeding service quality and leading to a 12 percent drop in ridership. As this development was reported to the market, Greyhound's stock tumbled nearly 25 percent. Over the course of a few months, as problems with Trips continued to surface, the stock price dropped another 60 percent (Tomsho, 1994).

In 1999, a software glitch brought online auction pioneer eBay's network to its knees for three days in a row, costing the company an estimated $2 million a day (Li 1999). Frustrated customers flooded Internet bulletin boards with threats to leave the service. Investors sent eBay stock tumbling almost 10% with more than twice the normal trading volume in the third day of outages.

In February 2001, sports apparel giant Nike informed stock analysts that due to problems in the implementation of a new supply chain system, the company was unable to meet their third-quarter earnings estimate. The next day, Nike stock lost 20 percent of its value (CNET News, 2001; Worthen, 2002).

Research on runaway projects and information technology (IT) failure has often been motivated by the financial costs of failure and the loss in effectiveness and competitiveness resulting from failures (Keil, 1995; Lee and Myers, 2004; Lyytinen and Hirschheim, 1987). Research studies and industry reports have repeatedly pointed to the high frequency of IT failures as well as the financial burden they constitute (Goldfinch, 2007; Johnson, 1995; Standish Group, 2004). Failure rates are often estimated at 30 percent and the frequency of troubled projects at 46-53 percent (Goldfinch, 2007; Johnson, 1995; Standish Group, 2001).

One type of financial consequence that has not been considered in IT failure research is the potential loss in firm value that might result from IT failures as they become known to investors. Investors, as owners, are atop the corporate governance structure and their reaction to IT failures is not only a measure of the value loss from IT failure but also

a measure of the importance assigned to IT failures by a key stakeholder group. Therefore, investor reactions to IT failures can also be seen as an indication of whether owners are likely to exercise influence over corporate boards and top executives with regard to IT governance. Should the market be tolerant to IT failures, this could suggest a lack of pressure from owners to bring down the frequency and severity of IT failures. On the other hand, should the market indeed care, it would suggest an additional motivation for executives and IT professionals alike to improve corporate capabilities in deploying and operating IT.

While the literature on IT failure has devoted much attention to the roles of various stakeholders (Larsen and Myers, 1999; Mitev, 2000; Sauer, 1993; Wilson and Howcroft, 2005), the role of investors has received scant attention. Coverage of market value loss from IT failures in the practitioner press (Koch, 2004; Wailgum, 2008) and anecdotal evidence (such as the Greyhound, eBay and Nike examples above) provide further impetus for this study.

Using event study methodology, we provide in this paper the first systematic, empirical evidence of the impact that IT failures can have on the market value of a firm. In so doing, we contribute to the literature on IT failure as well as to the body of literature on strategic management and the impact of IT.

Event studies examining stock market reactions to announcements involving strategic technology decisions have reported positive value relevance of announcements such as decisions to make investments in IT resources (Dos Santos, Peffers, and Mauer, 1993), creation of new Chief Information Officer positions (Chatterjee, Richardson, and Zmud, 2001), and acquisitions of Internet firms (Uhlenbruck, Hitt, and Semadeni, 2006).

However, we found no event study dealing with the value impacts on the level of specific information systems, whether systems in operation or systems being implemented. Furthermore, prior IT-related event studies all concern announcements about decisions or actions that *may* come to have business impacts at some later point in time (Dos Santos et al., 1993; Hunter, 2003; Koh and Venkatraman, 1991). Announcements of positive initiatives require of the market to assess the expected value of benefits resulting from the initiative as well as the probability that these positive results will actually materialize. Perhaps because of this, some prior studies have shown mixed results, for example identifying significant market valuation effects only for subsets of the sample (Dos Santos et al., 1993; Hunter, 2003). In contrast, IT failures are events that *have materialized* and the challenge for the market is to assess the expected loss in value from the failure. Thus, in contrast to previous IT event studies, this research examines the value impact of events that relate directly to actual shortcomings in the firm's IT capabilities and have direct consequences for business performance.

Previous studies use the resource-based view (RBV) (Barney, 1991; Teece, Pisano, and Shuen, 1997) to demonstrate the market value impact of IT-related announcements by associating IT capabilities and firm value. Following this lead, we are able both to add to existing knowledge about IT failures and how they are assessed and to link IT failure studies to RBV-based studies of the business value of IT (Barua and Mukhopadhyay, 2000; Sambamurthy, Bharadwaj, and Grover, 2003; Wade and Hulland, 2004).

The underlying questions driving this study are: (1) How much do investors care about IT failures? and (2) Do circumstances under which failures occur impact the market's assessment? In other words, when corporate resources and capabilities fail unexpectedly,

how does this impact firm value and does the extent of value loss vary under different conditions? Our study helps provide answers to these questions.

BACKGROUND AND HYPOTHESES

A variety of aspects of IT failure have been covered in the literature. Typologies of failure have been developed, particularly for systems that fail during development (Bussen and Myers, 1997; Lucas, 1975; Lyytinen and Hirschheim, 1987). Factors that promote failure have been identified (Beynon-Davies, 1995; Bussen and Myers, 1997; Fitzgerald and Russo, 2005; Irani, Sharif, and Love, 2001) and processes and mechanisms that contribute to failure have been described (Myers, 1994; Sauer, 1993).

An underlying view of much of the IT failure literature is that failure is caused by shortcomings in knowledge, competencies and abilities concerning processes of IT governance, IS development, change management, IS implementation, IS operation and IS use (Fitzgerald and Russo, 2005; Irani et al., 2001; Keil, 1995; Markus and Keil, 1994; Sauer, 1993). IT failures have also been characterized as organizational failures (Goulielmos, 2003; 2005) and have been seen as having long-term consequences for organizational learning abilities (Lyytinen and Robey, 1999).[1] While the consequences of IT failures are often characterized as highly negative, few if any attempts have been made to quantify the consequences of failure for firms or their shareholders.

The view of IT failures as being caused by a deficit in organizational capabilities can be directly linked to the resource-based view of the firm. For example, studies within IS that employ RBV commonly include the following IS/IT capabilities related to

[1] It can also be noted that the study, concepts and phenomena of IS success (DeLone and McLean, 1992; 2003) and IS failure are not directly comparable opposites. For a short discussion on similarities and differences between IS success and IS failure, please see Appendix A.

avoidance of IT failures: capability to manage IT-related change, practices and capabilities for information processing, ability to experiment with new technologies and conduct technical innovation, and ability to assimilate IT resources into organizational processes (Wade and Hulland, 2004).

The resource based view portrays firms as heterogeneous bundles of idiosyncratic resources and capabilities (Barney, 1991). Capabilities are "information-based tangible or intangible processes that are firm-specific and developed over time through complex interactions among the firm's resources" (Amit and Schoemaker, 1993, p. 35). Using the rubric of RBV, a firm's IT capabilities have been described as valuable business resources that underpin its competitive position by improving operational efficiency (Litan and Rivlin, 2001) and enabling key strategic initiatives such as customer relationship management, supply chain integration, and business process integration (Ray, Barney, and Muhanna, 2004). Numerous studies have used RBV as a theoretical lens for examining how firms are able to leverage IT resources and competencies (Amit and Zott, 2001; Gregor, Martin, Fernandez, Stern, and Vitale, 2006; Powell and Dent-Micallef, 1997; Ray et al., 2004; Rivard, Raymond, and Verreault, 2006; Tippins and Sohi, 2003). For example, Powell and Dent-Micallef (1997) showed how firms have outperformed their competition by using IT to leverage complementary human and business resources, Tippins and Sohi (2003) examined the role of organizational learning in mediating the effects of IT competency on firm performance, and Tarafdar and Gordon (2007) showed how specific IT competencies can affect the implementation of process innovations. At this point, it is well accepted that IT applications provide value only when they are successfully

implemented, adopted and used in day-to-day operations and decision-making (Markus and Keil, 1994).

Although IT systems are likely to differ in their strategic value to the organization, they nevertheless cause significant disruptions to a firm's business operations when they fail. The Greyhound Trips system failure described at the beginning of this article was one in which the failure of an IT system impeded the company's ability to issue tickets and led to significant delays for travelers, ultimately leading to a decline in ridership. Over and beyond the direct costs associated with fixing the crashed system and losses due to lower productivity, IT failures can result in indirect costs as well. As firms increasingly digitize their supplier and customer facing processes, the consequences of IT failure are greater and more visible than ever before, as illustrated by the Nike example at the beginning of this article. Failures adversely impact not only the firm's operations but also its external reputation and can result in investors questioning the capability of the firm to achieve growth and revenue targets. In the RBV sense, since the competitive advantage of a firm lies in its "…managerial and organizational processes shaped by specific asset positions" (Teece et al., 1997), IT failures can raise significant doubts about the quality of the underlying IT assets and the organizational processes that are enabled by these IT systems.

The resource-based view emphasizes that IT resources contribute to business performance through combination with other resources (Clemons and Row, 1991; Powell and Dent-Micallef, 1997), such as when information systems compliment highly effective business processes (Tarafdar and Gordon, 2007). For example, an off-the-shelf supply-chain system, while itself a commodity resource, can combine with existing supplier trust and relationships to form a mutually reinforcing and advantage-creating resource bundle.

Consequently, when IT systems fail, they weaken the strength and value of the complementary resource bundle. For example, supplier trust can erode and the entire supplier relationship network can be affected when supply chain systems cause operational disruptions. These in turn can impact the growth and revenue prospects of the firm, and as a result we expect investors on average to perceive the strong potential for negative returns due to IT failures. Thus, we state the following hypothesis:

H1: IT failures will result in negative abnormal returns for the firm.

Type of IT Failure: Operating vs. Implementation

In this study, we differentiate between two types of IT failures: *operating* failures that impact existing systems and *implementation* failures that involve new systems development projects.[2] From an RBV point of view (Eisenhardt and Martin, 2000; Teece et al., 1997), operating IT failures can be seen as evidence of shortcomings in the operational capabilities of the firm, the capabilities to execute processes that produce firm outputs. Similarly, implementation failures can be seen as resulting from shortcomings in the firm's dynamic capabilities, its capacity to adapt to changing conditions and to improve its functioning and performance (Eisenhardt and Martin, 2000; Teece et al., 1997; Winter, 2003). We posit that both types of IT failures can result in business losses and can therefore be expected to affect earnings.

[2] When reviewing existing notions of IT failure we found that they focus primarily on breakdowns that occur during the development process, involving systems that do not meet specifications, are not developed on schedule or within budget, are delivered but not used, or fail to meet stakeholder expectations (Lyytinen 1987). Such notions of failure center on IT project implementation and virtually ignore failures that can disrupt existing systems and infrastructure. Our coding scheme, which differentiates between implementation failures and operating failures was inspired by the real options literature which distinguishes between operating options and growth options (Benaroch, 2002), as well as by the RBV literature which distinguishes between operational capabilities (related to business performance) and dynamic capabilities (related to business improvement) (Eisenhardt and Martin, 2000; Teece et al., 1997).

Operating failures, or failures of existing systems, represent temporary *glitches* that can lead to business disruptions. When such failures are unexpected (e.g., a hardware crash), they can cause significant havoc, as in the example of eBay above, in which the company's servers crashed, causing $2 million a day in losses to the company (Li, 1999). While such failures can also signal a weaker capability in IT operations (Wade and Hulland, 2004), they are however, less likely to have long-term implications since normal business operations often resume when the systems are restored.

In contrast to maintaining operational systems, the implementation of information systems requires that firms possess dynamic capabilities related to experimenting with new technologies (Jarvenpaa and Leidner, 1998) and bundling them creatively with other resources into effective business processes (Rivard et al., 2006; Tarafdar and Gordon, 2007). Consequently, when news about an IT implementation failure become public information, this sends a signal to the market that, in RBV terms, the firm does not have the dynamic capabilities required to successfully make use of a technology resource in order to improve business processes.

Investments in new IT systems represent growth options for the firm (Benaroch, 2002) and therefore implementation failures such as the ones experienced by Greyhound and Nike often have both short-term and long-term implications. In the short-term, a firm may incur additional costs to turn-around the troubled project (Keil and Robey, 1999) or to write it off. Over the long-term, an implementation failure may delay critical infrastructure, thereby eroding the value of a firm's growth options. In the case of strategic information systems, such failures could prevent a firm from entering new markets or offering new products and services (Litan and Rivlin, 2001). An implementation failure

thus indicates that a firm's future competitiveness (which is based on its core capabilities)—and by consequence the firm's future earnings—can be put into question.

Since operating failures involving existing systems are not expected to hamper a firm's growth options, whereas implementation failures involving new systems may reduce the value of a firm's growth options and its expected future earnings, we expect that the type of failure will be significant in how the market reacts. Specifically, we state the following hypothesis:

> *H2: The negative abnormal returns due to implementation (new system) failures will be larger when compared to operating (existing system) failures.*

Severity of IT Failure

Regardless of the type, IT failures can differ significantly in the extent of damage they cause to the firm's operations, growth potential and earnings capacity. For example, failures of strategically important systems (Kanter, 2001; Litan and Rivlin, 2001) and failures of systems on which the business is operationally dependent (Mähring and Keil, 2008; Montealegre and Keil, 2000) are generally likely to yield more severe consequences than failures involving systems that are not critical for production or service delivery. Further, when existing systems or new system implementations fail, they can cause disruptions for both the firm and its customers, as seen in the Greyhound and eBay examples given earlier. In the eBay incident, the severity of the outage impeded it from doing business and shook investors' faith in the company. Severity of failure is conceived here as the extent to which the failure causes a minor or major disruption in customer transactions (a proxy for the strategic importance of the system to the firm) and whether the scope of the failure is narrow or broad (i.e., number of systems or people affected).

In the resource-based view, a more severe IT failure, such as a complete network outage for a telecommunications carrier, provides a stronger indication that underlying capabilities are deficient. Conversely, a less severe failure, such as a failed deployment of a beta release in a small test market, constitutes a weaker indicator of IT capability deficiencies. Furthermore, more severe failures indicate deficiencies in both IT capabilities and managerial capabilities (Mahoney and Pandian, 1992; Penrose, 1959). Managerial capabilities are linked to the firm's capacity for dealing with contingencies, prioritizing, making judgment calls and adapting to change (Adner and Helfat, 2003; Castanias and Helfat, 1991)—in other words, getting the most important things right and fixing problems effectively when they occur. Consequently, firms with well-developed managerial and IT capabilities (Bharadwaj, 2000; Powell and Dent-Micallef, 1997; Wade and Hulland, 2004) will be much less likely to mismanage a full-scale implementation of a mission-critical system, or neglect maintenance of such systems to the point of large-scale failure events occurring. Based on this argument, we suggest the following hypothesis:

H3: The negative abnormal returns will be larger for IT failures of greater severity.

History of IT Failures

In her analysis of organizational failures, Vaughan (1999) characterizes them as events that deviate from both formal design goals and normative standards either in the fact of their occurrence or in their consequences. While some failures might result purely from chance (e.g. an unexpected virus attack on an IT system), the management literature suggests that many failures occur because of the characteristics of the organizational system (Durkheim, 1966; Vaughan, 1999). Firm characteristics such as structure, processes, and problem-solving routines interact to produce failures (Elsbach and Sutton,

1992). Thus, repeated failures in a firm may be symptomatic of organizational pathologies, such as a tendency to engage in escalation of commitment to failing courses of action (Staw, 1997).

Extending this logic to IT systems, it may be argued that the conditions for repeated IT failures are often embedded in the firm's IT operations and its learning abilities. Since capabilities are evidenced in repeated patterns of action (Zollo and Winter, 2002), repeated failures that pertain to the same set of capabilities provide stronger and more certain evidence of shortcomings in operational and/or dynamic capabilities.

Lyytinen and Robey (1999) investigated a culture of organizational IT failure and examined why certain IS organizations are prone to repeated failures in their systems development practices. Among the major reasons cited for repeated failure are poor software development and ineffective project management practices that have persisted over extended time periods and made IS departments impervious to change. Other factors such as limits on organizational intelligence, disincentives for learning, and organizational designs that inhibit information and knowledge sharing also contribute to repeated failures in IS projects (Ciborra and Lanzara, 1994; Lyytinen, 1987). In the resource-based view, these reasons taken together translate into inadequate dynamic capabilities (Eisenhardt and Martin, 2000; Zollo and Winter, 2002), the capabilities associated with the organization's ability to learn and change, including the ability to question and change established procedures and modus operandi in a manner akin to double-loop learning (Argyris and Schön, 1978). Repeated IT failures thus serve as signals to the investment community of inherent deficiencies in capabilities that cause repetitive breakdown of key IT systems. Furthermore, they are indicative of the firm's inability to develop crucial core capabilities

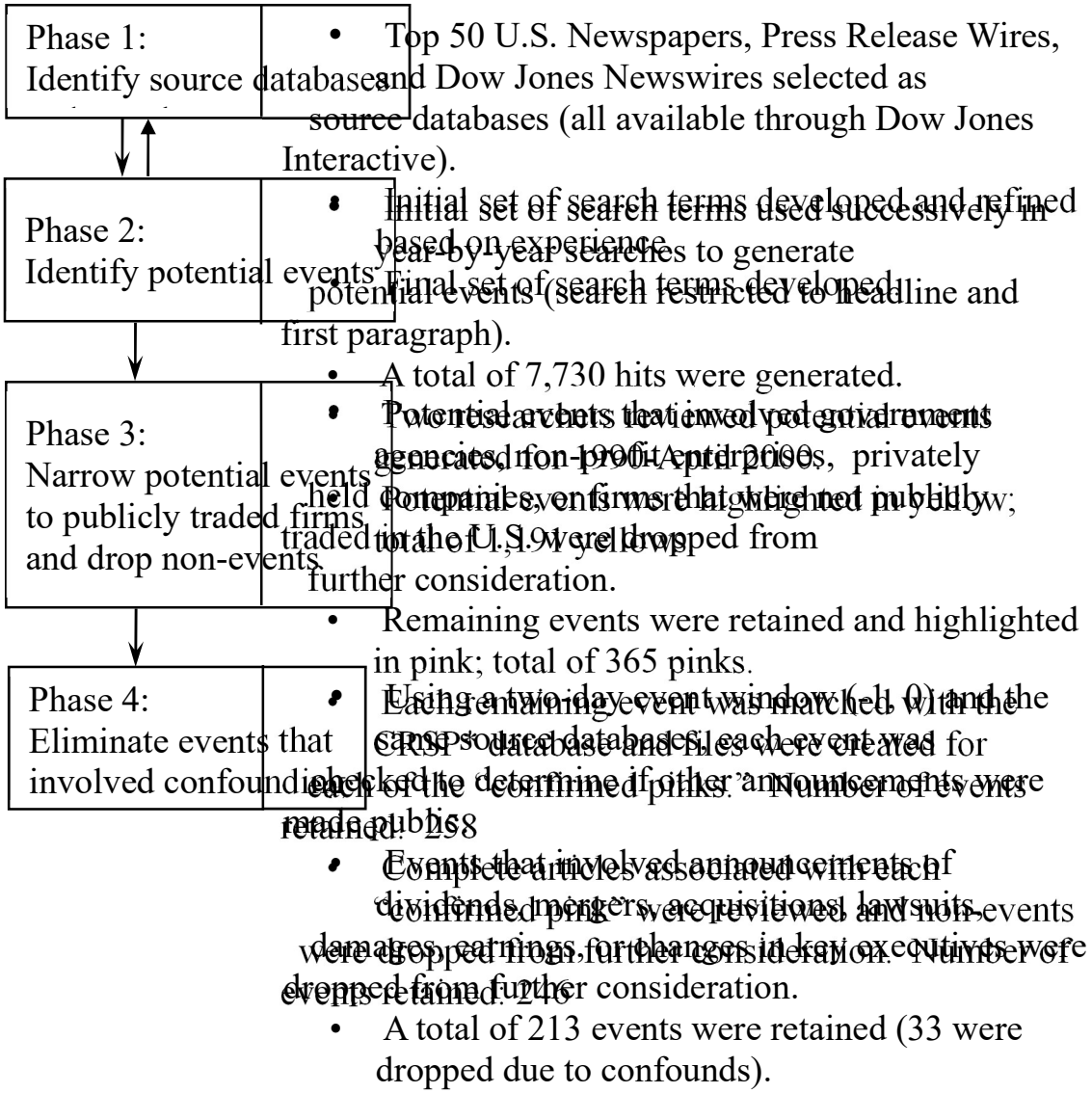
through learning from past failures (Andreu and Ciborra, 1996). Finally, in the case of repeated operating failures, they also signal the need for new investments targeted at improving the robustness and reliability of IT systems. This leads to our final hypothesis:

H4: Negative abnormal return will be larger for firms that experience repeated IT failures

RESEARCH APPROACH

The hypothesized relationships were tested on a sample of IT failure events constructed using a four-phase process as depicted in Figure 1. Figure 2 shows the seventeen search terms used to identify the failure events.

Figure 1. Sample Selection Methodology



*The Center for Research in Security Prices.

Figure 2. List of Search Terms*

• Software fail$ • Software project fail$ • Information system fail$ • Information technology fail$ • Computer fail$ • Computer bug	• Enterprise resource planning fail$ • ERP fail$ • Computer glitch • Computer problem • Software snafu • Web glitch	• Software glitch • Software problem • Software error • Denial of service attack$ • Software bug

*The dollar sign used at the end of some search terms was used as a wildcard to give greater flexibility in the search process; ERP fail$ for example would capture "ERP failure," "ERP failed," and related variants.

Event coding

The failure events were subjected to content analysis and coded for failure type and failure severity. Events were coded independently by two of the authors as either operating failures (i.e., failures to existing systems due to hardware or software problems) or implementation failures (i.e., failures of new system implementation projects). An ICC of 0.96 was obtained indicating very high inter-rater reliability. In those few instances in which there was not agreement, the two coders discussed the event and were able to reach a consensus on the most appropriate code.

Next, events were coded as *low, medium,* or *high* severity.[3] This coding scheme is consistent with the approach commonly taken in medicine when researchers seek to

[3] There was no literature that we could identify that was informative for coding failure severity; hence we created our own systematic approach. Using the concepts of failure scope and duration, we started with an 8-level coding scheme for assessing failure severity that took into account whether the failure represented a *minor* or *major* disruption in customer transactions (a proxy for the strategic importance of the system to the firm), whether the scope of the failure (i.e., number of systems or people affected) was *narrow* or *broad, and* whether the failure was *short lived* (defined as under a half-hour) or *long-lived*. Failure duration proved to be somewhat problematic as many articles did not report failure duration and the cut-off that we had originally established for short vs. long duration seemed somewhat arbitrary. Hence, a decision was made to collapse the coding scheme to something simpler. Nevertheless, the original scheme allowed us to develop a systematic approach for coding these events and insured that the two coders were considering the same set of factors. This improved the reliability of the coding. In order to collapse the original coding scheme into the three categories mentioned earlier, events (short or long-lived) that posed a minor disruption and had narrow scope were coded as low severity events. Events (short or long-lived) that posed a major disruption and had broad scope were coded as high severity events. All other events were coded as medium severity.

understand adverse drug reactions. Published papers in this area have coded adverse reactions as mild, moderate, or severe (see, for example, Murphy, Kremer, Rodrigues, and Schatzberg, 2003). Events were coded independently as before and an ICC of 0.93 was obtained indicating very high inter-rater reliability.

Control Variables

Firm and industry characteristics are important controls in this type of analysis. Larger firms with sufficient organizational slack may be less affected by a failure as they can quickly redeploy back-up resources. Firms in high-growth mode are more likely to be critically reliant on their IT systems for expansion and therefore more likely to be impacted by failures. Finally, we expect the impact of failures to be more pronounced for IT intensive industries such as telecom and financial services than for less IT intensive industries such as mining. We accounted for these differences by introducing appropriate controls for firm size, firm growth rate, and industry IT intensiveness. To further control for industry, a second control was introduced using the more traditional manufacturing/services categorization. The study variables are described in Figure 3.

Event Study Methodology

We used the event study methodology to estimate the abnormal stock price effects associated with IT failures. The methodology is based on the market model of daily stock price returns, which is used to estimate the abnormal return (AR) for stock *i* on day *t*, as $AR_{it} = R_{it} - (\alpha_i + \beta_i R_{mt})$[4].

[4] R_{it} is the return of stock *i* on day *t*; R_{mt} is the return of the market portfolio on day *t* using the S&P 500 as the index of the market; α_i is the is the intercept of the relationship for stock *i* and β_i is the slope of the relationship for stock *i*.

Figure 3. Study Variables

Variable name	Control Variable?	Operationalization
Firm Size	Yes	Natural logarithm of the market value of firm equity at the end of the year immediately preceding the event date[5].
Firm Growth	Yes	Percentage increase in yearly sales determined as (sale(t) – sale(t-1))/sale(t), where t represents the most recent fiscal year ending prior to the event date.
Industry IT Intensiveness	Yes	Average IT spending per industry was used as a proxy for industry IT intensiveness. We used an indicator variable coded as "1" if the firm belonged to one of the top three industries for IT spending (telecommunications – SIC 4800 to 4899, financial services/investment firms, brokerages, dealers, etc. – SIC 6211 to 6799, and banking/credit institutions – SIC 6000 to 6159) and "0" otherwise.
Industry Type	Yes	The traditional manufacturing/services categorization was used to differentiate between industry types, coded as "1" for manufacturing firms and "0" for service firms.
Failure Type	No	Binary variable to differentiate implementation failures from operating failures, coded as "1" for implementation failures and "0" for operating failures.
Failure Severity	No	Coded as "1", "2", or "3" corresponding to low, medium, or high severity.
History of failures	No	Binary variable to differentiate firms with more than one failure during the study period, coded as "1" if the failure had been preceded by earlier failure(s) and "0" otherwise.

As explained in this section, we followed McWilliams and Siegel's (1997) recommendations to insure that the results of our event study analysis were robust. First, our sample size of 213 failure events over a 10 year period was large enough to guard against the possibility that a few outliers might be influencing the results.[6] Our data set was also pre-checked for influential outliers before we ran the regressions; no significant outliers were detected. Second, in keeping with the recommendation of using the smallest justifiable event window, we opted for a narrow two-day event window comprising the event day and the day prior to it, to account for the possibility that information about the

[5] Three alternate proxies for size, namely, total number of employees, total sales, and total asset value were also used to test the robustness of the results.

[6] We chose the time span 1990-2000 to obtain a large sample size over an extended time period, while at the same time avoiding the potential problems associated with using data from the period of the crash of the dot-com bubble (2001-2002).

failure might have reached the market the day prior to the actual report in the newspapers. In order to estimate abnormal returns, we used an estimation window of 120 trading days prior to the event and ending 2 days before the event (-120 to -2). Two alternate estimation windows of -200 to -50 and -250 to -50 were also used to verify the robustness of the results. The daily abnormal returns were averaged over the sample of 213 announcements and over the two days (t=-1,0) of the event period to examine if the mean cumulative abnormal return $CAR_{-1,0}$:

$$\overline{CAR_{-1,0}} = \frac{1}{213}\sum_{i=1}^{213}\sum_{t=-1}^{0} AR_{it}$$

was significantly negative over the event window. To test the hypothesis that the mean CAR over the event period was significantly different from zero, we used the student's *t*-test, which takes the form:

$t =$ 

Generalized least square (GLS) regression analysis was used to test the remaining hypotheses (H2-H5) pertaining to failure type, failure severity, and firm's history of failures, with the remaining variables (size, firm-growth, industry IT intensiveness, and industry type) included as controls:

$$CAR_{-1,0} = \beta_0 + \beta_1 firmsize + \beta_2 firmgrowth + \beta_3 industryITintensiveness + \beta_4 industrytype + \beta_5 failuretype + \beta_6 failureseverity + \beta_7 historyoffailure + \varepsilon$$

The reason for using generalized least square regression analysis instead of standard multivariate regression analysis was that 36 of the firms in our sample experienced multiple failures over the analysis time period, which raises the issue of non-independence of observations. The use of GLS regression corrects for this potential problem because it

allows for the modeling of within-sample dependencies, provides correct standard errors and ensures that firms providing more data (i.e., with multiple failure events) do not unduly influence the results. Moreover, we used clustered robust standard errors based on the Huber-White sandwich estimator and thus corrected for non-independence of observations impacting the standard errors used.

RESULTS

Summary statistics are shown in Table 1, distribution of firms across industries in Table 2 and a correlation matrix in Table 3.

Table 1: Summary Statistics

Variable	Observations	Mean	Std.Dev.	Min	Max
CAR	206	-0.022	0.060	-0.315	0.092
Failure type *(H2)*	206	0.16	0.37	0	1
Failure severity *(H3)*	206	1.985	0.774	1	3
Failure history *(H4)*	211	0.431	0.496	0	1
Firm size	211	9.024	1.883	4.005	13.041
Firm growth	206	0.165	0.230	-0.579	0.893
Industry IT intensity	211	0.564	0. 497	0	1
Industry type	211	0.227	0.420	0	1

Table 2: Distribution of Firms in Sample across Industries

Industry	Percentage of Sample
Financial and real estate services	24%
Equipment manufacturing	23%
Communication services	17%
Online and media services	11%
Retail	9%
Basic manufacturing	8%
Transportation services	6%
Utilities	2%
Professional services	2%
Total	100%

Table 3: Correlation Matrix[7]

	CAR	Failure type *(H2)*	Failure severity *(H3)*	Failure history *(H4)*	Firm size	Firm growth	Industry IT intensity	Industry type
CAR	1.000							
Failure type *(H2)*	-0.187**	1.000						
Failure severity *(H3)*	-0.354***	0.127	1.000					
Failure history *(H4)*	0.020	-0.226***	-0.034	1.000				
Firm size	0.385***	-0.240***	-0.239***	0.371***	1.000			
Firm growth	-0.190**	-0.051	0.184**	0.067	-0.017	1.000		
Industry IT intensity	0.054	-0.213**	-0.029	0.187**	0.168*	-0.024	1.000	
Industry type	-0.062	0.223**	-0.005	-0.313***	-0.175*	-0.262***	-0.617***	1.000

*** $p<0.001$; ** $p<0.01$; * $p<0.05$

[7] With one exception, the correlations are all in the range of -0.26-0.37. Subsequent analysis revealed that the variance inflation factors were substantially below the critical value of ten, indicating that multicollinearity was not an issue (Neter, Wasserman, and Kutner, 1995).

Table 4 (Panel A) shows the mean and median abnormal returns for the days $t = -1$, $t = 0$ and $t = (-1,0)$ using two different estimation windows[8]. Abnormal returns were significantly negative on both days. On day -1, there was a 0.9% average drop in stock returns and on day 0, a 1.1% average drop in stock returns giving a combined drop of 2% over the event period (t = -4.96).

Panel B (Table 4) shows results of the GLS regression analysis. The overall R^2 was 25% both for the -120 to -2 estimation window as well as the -200 to -50 estimation window and the same pattern of results held across the two estimation windows. All of the hypotheses were supported. Failure type (H2) was significant at the $p<0.01$ level, and the sign was consistent with our expectation that implementation failures would be more damaging than operating failures to firm value. Failure severity (H3) was significant at the $p<0.01$ level, indicating that more severe failures had a greater influence on the negative CAR. A firm's prior history of IT failures (H4) was also significant at the $p<0.01$ level, showing that subsequent failures by the same firm extracted a larger drop in share prices. Among the control variables, firm size was significant ($p<0.01$) which is in line with findings from other event studies that have shown that small firms tend to experience a greater drop in value due to unexpected events. Industry type was also found to be significant ($p<0.10$), suggesting that the market penalizes manufacturing firms that incur failures slightly more than service firms.

[8] Results obtained for an even longer estimation window of -250 to -50 showed remarkable consistency to the results reported here.

Table 4. Event Study Results

PANEL A						
Measures	**Estimation Window (-120 to -2)**			**Estimation Window (-200 to -50)**		
	Day -1	***Day 0***	***Days -1,0***	***Day -1***	***Day 0***	***Days -1,0***
# of Events	213	213	213	206	206	206
Mean Abnormal Return	**-0.009**	**-0.011**	**-0.020**	**-0.01**	**-0.01**	**-0.02**
Std. Dev of Return	0.034	0.040	0.060	0.034	0.04	0.06
Median Abnormal Return	-0.005	-0.003	-0.009	-0.004	-0.004	-0.001
Student's T *(p-value)*	-3.90 *(0.0001)*	-4.06 *(<0.0001)*	-4.96 *(<0.0001)*	-4.22 *(<0.0001)*	-4.38 *(<0.0001)*	-5.32 *(<0.0001)*
Sign Test *(p-value)*	-19 *(0.011)*	-18 *(0.016)*	-30 *(<0.0001)*	-27.5 *(0.0002)*	-15 (0.04)	-32 *(<0.0001)*
Signed rank *(p-value)*	-3516 *<0.0001*	-2867 *0.00*	-4105 *(<0.0001)*	-3536 *(<0.0001)*	-3122.5 (0.0002)	-4466.5 *(<0.0001)*

PANEL B				
Variable	**Estimation Window (-120 to -2)**		**Estimation Window (-200 to -50)**	
	Standardized Estimate	***t*-value**	**Standardized estimate**	***t*-value**
Intercept	-0.08	-2.59***	-0.08	-2.59***
Failure type (*H2)*	-0.02	-2.60***	-0.02	-2.63***
Failure severity (*H3*)	-0.01	-3.56***	-0.01	-3.51***
Failure history (*H4)*	-0.03	-4.02***	-0.03	-4.10***
Controls				
Firm size	0.01	3.62***	0.01	3.64***
Firm growth	-0.02	-1.07	-0.03	-1.15
Industry IT intensity	-0.01	-1.08	-0.01	-1.14
Industry type	-0.02	-1.43*	-0.02	-1.48*
Overall R^2	0.2509		0.2507	

**** p<0.01; ** p<0.05; * p<0.10; one-tailed test of significance*

In order to assess the robustness of the results, a number of alternate analyses were performed with variations in control variables: In addition to the top three industries by IT spending, we ran separate analyses including the top five industries for IT spending (adding insurance and retail). We then divided the sample using the more traditional manufacturing/services categorization. We ran three different regressions using these controls, one at a time and then both simultaneously. No significant changes in results were observed for either analysis. We report the model that uses both IT intensiveness and Industry (Manufacturing/Services) as controls. In addition, since it could be conceived that

a correlation would exist between failure type and failure severity, we checked for interaction effects between these control variables. This did not produce any significant results.

As described above, we used GLS regression to correct for the firms in our sample that experienced multiple failures over the analysis time period; we also used clustered robust standard errors based to correct for non-independence of observations impacting the standard errors used. To test that these measures were effective, we reviewed the data set and identified 29 instances of potentially confounding events within our primary estimation window and 35 instances within our alternative estimation window. All these instances were removed from the sample and the analysis was rerun, producing essentially the same result. We thus concluded that the risk of bias from non-independence had been appropriately addressed.

Our model uses market value as a proxy for firm size. To check the robustness of our findings, we also examined three alternate proxies for size, namely, total number of employees, total sales, and total asset value. The result pattern obtained was similar regardless of which proxy was used for firm size.

Finally, some recent research suggests that investors might take time to react to news (Chan, 2003). Consequently, events taking place prior to the event window may still be confounding events.[9] However, we used two alternate estimation windows (-200 to -50, and -250 to -50) in addition to the primary estimation window (-120 to -2), and received the same basic pattern of results regardless of the estimation window used. This suggests that we can be reasonably confident that the market's reaction to prior events is not biasing

[9] We thank one anonymous reviewer for pointing us to this potential risk.

the results of our study.

DISCUSSION AND IMPLICATIONS

We now discuss the results of our study and its implications, using the two guiding research questions as an organizing framework.

RQ1: How much to investors care about IT failures?

The strategic management literature has seen a growing interest in understanding the complex relationship between IT and business value. To date, these studies have drawn on RBV to show how IT-based resources can be leveraged to improve firm performance. Previous event studies in this area have examined positive stock market reactions to *announcements* of strategic IT investment decisions (Lee, 2001; Uhlenbruck et al., 2006). Our study contributes to this growing body of literature, as well as to the study of IT failure, by examining the value loss associated with *actual* IT failures, and thus provides a different angle for testing the tenets of RBV. The results of this research show that when IT resources fail unexpectedly, they most definitely have a negative impact on firm value. Indeed, our results show that there is, on average, nearly a 2% decline in market value when firms experience IT failures. This average is greater than the corresponding positive market value effect observed in studies of announcements concerning intra-firm IT investments (Dardan, Stylianou, and Kumar, 2006; Dos Santos et al., 1993; Oh, Kim, and Richardson, 2006), with CAR values ranging from 0.09% to 0.36%, and up to 1.03% for a sub-sample).

By demonstrating that operational and implementation failures of IT systems negatively impact firm value, our study puts the spotlight on an important stakeholder

category, investors, which is often neglected in IT failure studies. In short, the evidence clearly indicates that investors do care about IT failures.

RQ2: Do failure circumstances impact the market's assessment?

Our results also show that the market's assessment takes into account the circumstances under which failure occurs. For example, investors were found to respond more negatively to implementation failures than to operating failures. Further, the study provides evidence that more severe IT failures extract a greater penalty from investors, and that firms that have a history of IT failures (i.e., those that had multiple failures during the time period) tend to suffer greater negative impact.

These results suggest that, from an RBV perspective, investors behave quite rationally in their assessment of IT failures. Implementation failures cause operational problems and indicate deficits in IT deployment capability. Since these failures indicate shortcomings in dynamic capabilities that put the firm's future earnings capacity in question, it is logical that investors react strongly to news about implementation failures.[10] At the same time, operating failures represent more than just a temporary glitch; they are manifestations of underlying operational capability deficiencies that cannot simply be neglected in the assessment of firm value. Similarly, repeated failures provide stronger evidence of capability deficiencies and consequently the investment community extracts a greater penalty from firms that have failed to learn from past mistakes. Finally, more severe

[10] An alternative explanation for the difference in value loss would be that the greater value drop for implementation failures partly or completely results from a market correction of an earlier abnormal gain following a positive announcement about the same initiative. In order for this to the case, we would expect to find prior press coverage for a large fraction of our implementation failure events. However, we found that very few of the implementation failures in the study had received press coverage prior to the point of failure. Thus, while theoretically plausible, this alternative explanation does not seem to be compelling.

failures can hamper the firm's operations and future growth options to a greater extent than less severe failures. In essence, our findings indicate that the market cares enough and knows enough to assess IT failures differently depending on their nature and circumstances.

Implications for Practice

The fact that the market pays attention to IT failures can also provide useful input to research on IT governance and executive involvement in IT. Specifically, our results suggest that corporate boards and executives could be motivated to engage in IT governance activities by the market value impact of IT-related events that become known to the market. It should be of interest to study if attention from owners can, over time, make executives pay closer attention to the state of the firm's existing systems as well as current development projects.

Presumably, executives, particularly those holding stock options or company shares in their retirement plans, would not like to see a 2% in market value over a two-day period. Yet, this is precisely what they will see on average when their firm experiences an IT failure. Indeed, the results of our study suggest that when a firm's market value is negatively impacted by an IT failure, the average market value loss amounts to $490 million over a two-day event window.[11] Such a significant market reaction to IT failures underscores the importance of ensuring the reliability of existing IT systems as well as avoiding costly

[11] Not all firms in our sample registered a negative abnormal return subsequent to the failure. While the sample was carefully weeded for confounding announcements, these were carried out at the firm level. A possible explanation for remaining positive CAR values in our sample is that on some trading days, all firms in an industry are lifted up by market factors. Probing the specifics for each event in our sample would require a case study type analysis that is beyond the scope of the current paper, which finds that *on average* firms suffer market losses due to IT failures.

implementation failures. The demonstrated negative market reaction to IT failures should help managers make the case for organizational changes needed to improve the reliability of existing IT systems and to ensure the success of new IT system implementations. It should also provide motivation for boards and executives to ensure that corporate IS/IT strategies and related IT governance practices—including project governance practices, regular assessment of core systems, business continuity risk analyses, and disaster recovery procedures—are appropriate, proactive and effective in minimizing the occurrence of IT failures. It is also important that these aspects, which all relate to ensuring business continuity, business agility and business improvement, are well covered in the firm's IS/IT strategy.

While executives should constantly reassess their firm's IT capabilities and take appropriate steps to reduce the probability of failures, they also need to proactively mitigate the impact of failures when they do occur. Knowing that stock markets will react negatively to reports of IT failures, it would be advisable for management to be proactive and engage in damage control to allay investor concerns during IT failures. This means that not only internal but also external impressions need to be carefully and proactively managed when closing down a failed project (Iacovou and Dexter, 1996; Montealegre and Keil, 2000).

Results from our study could be used by executives, chief information officers and other organizational actors as a justification for greater investments that can increase the robustness and reliability of a firm's core IT systems. Executives should also be aware that investors are more understanding when it comes to unexpected failures of existing systems than they are with respect to failures involving new implementations. Our data thus suggest

that from an investor relations standpoint executives should be particularly sensitive to the need to successfully manage new system implementations. Executives should also keep in mind the old adage: "Fool me once, shame on you! Fool me twice shame on me!" Our results clearly indicate that the when firms face multiple failures, the investment community is not willing to play the fool and expects them to learn from their mistakes and take the necessary precautions to prevent future failures.

Limitations and Directions for Future Research

As is common across all event studies, this research makes a number of standard assumptions. First, is the assumption of market efficiency, which implies that the release of new information about a firm is immediately reflected in its share price. Second, the study assumes that initial newspaper reports of IT failures represented "surprises" to the market. It is possible that in certain cases—for example, large implementation failures—some information about the impending failure leaked to the market prior to the announcement. Another possible limitation is that the market may react to negative news more severely than is warranted, leading to exaggerated estimates of the impact (Arrow, 1982; DeBondt and Makhija, 1988).

As noted earlier, we have followed McWilliams and Siegel's (1997) recommendations to avoid some of the pitfalls that can occur with event studies. By collecting our sample over a 10-year period and having a robust sample size of 213 events, we have addressed the most common limitation of event studies, which has to do with small samples and the possibility that a few outliers might be influencing the results. By keeping the smallest justifiable event window, we avoided possible confounding events that could have influenced share prices. We also demonstrated the robustness of our results by using

multiple estimation windows. Finally, a number of alternate analyses with varying control variables and exclusion of potentially confounding events were performed to further test for result robustness, with encouraging results.

As discussed earlier, the 2% decline in market value that we observed following an IT failure is greater than the positive market value effect observed in other studies of announcements concerning intra-firm IT investments. However, more research is needed to test whether this result holds up for positive and negative events concerning the same specified type of IT resource, drawn from the same sources spanning the same time period.[12] Should this be the case, it would suggest that market reactions to positive and negative events are different.

We have also shown that the market responds more negatively to implementation failures than to operating failures. It should also be noted that the market's attention to operating failures as such is interesting, since the IT failure literature is largely focused on implementation failures (e.g., Fitzgerald and Russo, 2005; Lucas, 1975; Myers, 1994; Sauer, 1993). Our findings suggest that further investigation of operating IT failures is warranted. In order to help balance attention to operating as well as implementation failures, a business continuity lens could prove useful. The institutional theory perspective of business continuity planning, (Zsidisin, Melnyk, and Ragatz, 2005) represents one such approach that has recently been employed in the supply chain literature with promising results. Further research could provide a finer grained analysis of particular types of implementation failures and operating failures and their relative impact on firm value.

While this study demonstrates that the decline in firm value is greater for more

[12] We thank the senior editor and one anonymous reviewer for pointing us to this interesting topic for further research.

severe IT failures, additional research is needed to develop and test alternative measures of failure severity and to discern what influences failure severity the most, from the perspective of the investor. There are obviously many potential factors that can affect the severity of an event and there is currently no information to suggest what the relative weights of these factors are.

Finally, our results indicate that firms with a history of IT failures exhibit a greater decline in firm value when an IT failure occurs. Future research should be directed toward understanding the response pattern to multiple failures in a more nuanced way, so that we may determine the exact nature of the relationship between the number of failures a firm has experienced in the past and be able to predict the penalty that will be associated with the next failure.

Several other research questions could be posed based on the findings reported here. Does the market extract equivalent penalties for failures involving IT systems for internet-based firms versus more traditional, non-internet-based firms? Do IT failures that involve breach of sensitive data such as customer credit history and medical information cause larger losses? Does the market extract equivalent penalties for outsourced versus insourced IT system failures? A more fine-grained examination of IT failures can provide answers to these interesting questions. Turning to the issue of reporting of IT failures, have the requirements of the Sarbanes-Oxley Act (which include regulations on disclosure of operational risks, including IT risks) increased the openness of firms with regards to their IT failures? Conversely, does the common practice of tying executive compensation to stock prices reduce the propensity of firms to disclose IT failures? We recognize that this study represents a beginning and not an end to the investigation of how IT failures affect

firms. It is our hope that this study will encourage future research in this interesting and important area.

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APPENDIX A: ARE IS SUCCESS AND IS FAILURE MIRROR IMAGES?

A partial correspondence exists between studies that identify factors in IS development that produce either success (DeLone and McLean, 2003) or failure (Fowler and Horan, 2007). For example, the presence or absence of top management support, effectiveness in project management and user involvement have proven important in both sets of studies (e.g., Bussen and Myers, 1997; DeLone and McLean, 2003; Doherty and King, 2001). A few studies have made attempts to directly compare success and failure factors (Fitzgerald and Russo, 2005; Fowler and Horan, 2007; Liebowitz, 1999) and some studies have explored the dynamics that may reverse success into failure or vice versa (Irani et al., 2001; Larsen and Myers, 1999).

However, IT failure and IT success are not mirror images. While there are overlaps in factors that promote versus impede success in development and use, the dynamics of success and failure are different (Mähring and Keil, 2008; Montealegre and Keil, 2000; Myers, 1994), as are the consequences of success versus failure (Brooks, 1975; Lyytinen and Robey, 1999). Consider, for example, the case of ERP implementation: Successful ERP implementation can provide a new and better platform for running the business and for developing business processes (Broadbent, Weill, and Neo, 1999; Srivardhana and Pawlowski, 2007). As such, a successfully implemented and used ERP system can from an RBV perspective be seen as a new organizational (IT) resource (Peppard and Ward, 2004). This resource can over time become intertwined into organizational processes in a way that leads to improvement of organizational capabilities (Eisenhardt and Martin, 2000) in areas such as logistics, inventory management and production (Peppard and Ward, 2004; Srivardhana and Pawlowski, 2007; Wade and Hulland, 2004). Thus, consequences (effects, benefits) often appear gradually over a longer time period and are often difficult to attribute

directly to any one IT resource because effects from IT resources are indirect (Clemons and Row, 1991). In contrast, a failure in development, implementation and use of an ERP may lead to quite different consequences, because it may carry with it an immediate deficiency in organizational resources and a breakdown of operative capabilities (Hendricks and Singhal, 2003). In the cases of Hershey (Stedman, 1999) and FoxMeyer Drugs (Bulkeley, 1996), failed ERP implementations led to breakdowns in logistics and order fulfillment from which the firms struggled to recuperate, in the case of FoxMeyer even failing to do so. Consequently, we argue that from an RBV perspective, the organizational consequences of IT failure versus IT success can be significantly, qualitatively different.